

Lattices

$$U = \{1, 2, 3, 4\}.$$

$$S, T \subseteq U$$

$$\left. \begin{aligned} \text{lub}(S, T) &= S \cup T \\ \text{glb}(S, T) &= S \cap T \end{aligned} \right\}$$

$(P(U), \subseteq) \rightarrow$ power set of U is a lattice.

Complete Lattices.

- # Any lattice lub & glb of the finite subsets of the ~~poset~~ poset (pairs) should be part of lattice.
- # Any complete lattice the lub & glb of ~~both~~ both finite & infinite subsets should be part of the lattice.

E.g. $X \rightarrow$ infinite subset.
 $L = \mathcal{P}(X) = \mathcal{P}(\mathbb{N})$.

L be a collection of all ^{finite} subsets of X
 (X is ordered by $\subseteq$).

For any finite subset $A, B \in L$.

$$\text{lub} = A \cup B, \quad \text{glb} = A \cap B.$$

Both of the union & intersection are in L .

$\therefore L$ is a Lattice

? if L is a complete lattice?

$$T = \left\{ \underbrace{\{0\}}, \underbrace{\{0, 1\}}, \underbrace{\{0, 1, 2\}} \dots \right\}.$$

part of L .

upper bound of T : $\rightarrow \mathbb{N} \rightarrow$ infinite and
 therefore not part of L .
 $\therefore L$ is not a complete lattice.

Equivalence (Partitions)

Given a set A and an index set I let
 $\emptyset \neq A_i \subseteq A$ for each $i \in I$

Then $\{A_i\}_{i \in I}$ is a partition of A if

$$(a) \quad A = \bigcup_{i \in I} A_i$$

$$(b) \quad A_i \cap A_j = \emptyset \quad \begin{matrix} i, j \in I \\ i \neq j \end{matrix}$$

$\hookrightarrow$ set A is broken / partitioned
into disjoint groups.

E.g.

$$A = \{1, 2, 3, \dots, 10\}$$

$$(n) \quad A_1 = \{1, 2, 3, 4, 5\}, \quad A_2 = \{6, 7, 8, 9, 10\}$$

$$A_1 \cup A_2 = A$$

$$A_1 \cap A_2 = \emptyset$$

Defⁿ

Equivalence classes.

Let $\cong$ be an equivalence relation on the set A . For any $x \in A$ the equivalence class of x denoted by $[x]$ is defined as

$$[x] = \{y \in A \mid y \cong x\}.$$

E.g. $\cong$ on $\mathbb{Z}$ s.t. $x \cong y$ if $\underline{4 \mid (x-y)}$
 is $\cong$ an equivalence relation?

$\rightarrow \{4k \mid k \in \mathbb{Z}\}$
 $\rightarrow \{4k+1 \mid k \in \mathbb{Z}\}$
 $\rightarrow \{4k+2 \mid k \in \mathbb{Z}\}$
 $\rightarrow \{4k+3 \mid k \in \mathbb{Z}\}$

$[0] = \{ \dots, -8, -4, 0, 4, 8, 12, \dots \}$
 $[1] = \{ \dots, -7, -3, 1, 5, \dots \}$
 $[2] = \{ \dots, -6, -2, 2, 6, \dots \}$
 $[3] = \{ \dots, -5, -1, 3, 7, \dots \}$
 $[0] = [-8] = [-4]$

Theorem:

$\approx$ on set A , $x, y \in A$

(a) $x \in [x]$

(b) $x \approx y \iff [x] = [y]$

(c) $[x] = [y] \iff [x] \cap [y] = \Phi$

Reflexive ~~property~~ property

Given ① $x \approx y$ Forward.

② let $w \in [x]$

③ $w \approx x$

④ $w \approx x, x \approx y \Rightarrow w \approx y$ ⑤ $w \in [y]$

⑥ $[x] \subseteq [y]$

⑦ $x \approx y \Rightarrow y \approx x$

⑧ Let $t \in [y] \Rightarrow t \approx y$

⑨ $t \approx y \& y \approx x \Rightarrow t \approx x \Rightarrow [y] \subseteq [x]$

⑩ $\therefore [x] = [y]$

Backward: given $[x] = [y]$

① ~~$x \in [x]$~~ . $x \in [x]$: (reflexive).

② $x \in [y]$. (given).

$$\Rightarrow x \approx y.$$

Either $[x] = [y]$ or $[x] \cap [y] = \emptyset$

Path forward is contradiction.

Let us say ① $[x] \cap [y] \neq \emptyset$

② $v \in A$, $v \in [x]$ and $v \in [y]$

$$v \approx x, \quad v \approx y$$

$$\downarrow$$
$$x \approx v, \quad v \approx y$$

(Symmetric property)

$$x \approx y \Rightarrow [x] = [y].$$

transitive property.

$\hookrightarrow$ ~~scribble~~

- H.W.
- (a) Any equivalence relation $\cong$ on A induces a partition on A .
 - (b) Any partition of A gives rise to an equivalence relation $\cong$ on A .

Hint: Use the previous three sub-theorems.

$$\underline{\underline{x \cong y}} \rightarrow \boxed{x, y \in A; \text{ same cell.}}$$

Sizes of (infinite) sets.

Defⁿ

For any set, A & B the function $f: A \rightarrow B$ is called a one-to-one correspondence if it is

- (a) one-to-one
- (b) onto

} bijective function f.

$$A = \mathbb{Z}^+ \quad , \quad B = 2\mathbb{Z}^+ \\ = \{2k \mid k \in \mathbb{Z}^+\} \\ = \{2, 4, 6, \dots\}$$

$$\hookrightarrow f: A \rightarrow B \\ \underline{f(x) = 2x}$$

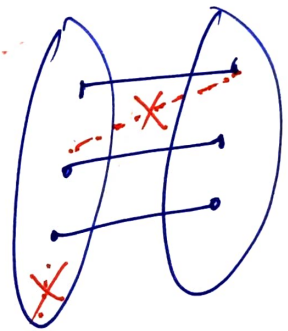
(a) one-to-one?
 $a_1, a_2 \in A$

Assume $f(a_1) = f(a_2)$

$$\therefore 2a_1 = 2a_2$$

$$\Rightarrow a_1 = a_2$$

(b) onto? $f(a) = 2a = b \in B$
 $a = b/2 \in A$



Defn

If A & B ~~are~~ are two sets

We say that they have the same sizes / cardinality ($A \sim B$)

if there exists a bijective mapping from $f: A \rightarrow B$.

(a) $A \sim A$

(b) If $A \sim B$, $B \sim A$

(c) If $A \sim B$ and $B \sim C$ then $A \sim C$.